Ch. 30 CW #1

Principles of Macroeconomics

February 27, 2026

1. Isabella takes $100 of currency from her wallet and deposits it into her checking account. If the bank adds the entire $100 to reserves, the money supply _____, but if the bank lends out some of the $100, the money supply _____.

   (a) increases; increases even more

   (b) increases; increases by less

   (c) is unchanged; increases

   (d) decreases; decreases by less

2. If the reserve ratio is $\frac{1}{4}$ and the central bank increases the quantity of reserves in the banking system by $120, the money supply increases by

   (a) $90.

   (b) $150.

   (c) $160.

   (d) $480.

3. Bill Murray deposits $500 that was in his sock drawer into a checking account at the local bank

   (a) How does the deposit initially (i.e. before any loans are made) change the T-account of the local bank? How does it change the money supply?

   (b) If banks maintain a reserve ratio of 10%, how will the local bank respond to the new deposit?

   (c) If every time the bank made a loan, the loan resulted in a new bank deposit equal to the amount of the loan, how much would the money supply in the economy expand in total?

   (d) If banks instead maintained a reserve ratio of 5%, how much would the money supply expand?

4. Dan Aykroyd withdraws $400 from his checking at the local bank and keeps it in his wallet.

   (a) If banks maintain a reserve ratio of 10%, how will the withdrawal affect the money supply?

   (b) If banks maintain a reserve ratio of 20%, how will the withdrawal affect the money supply?